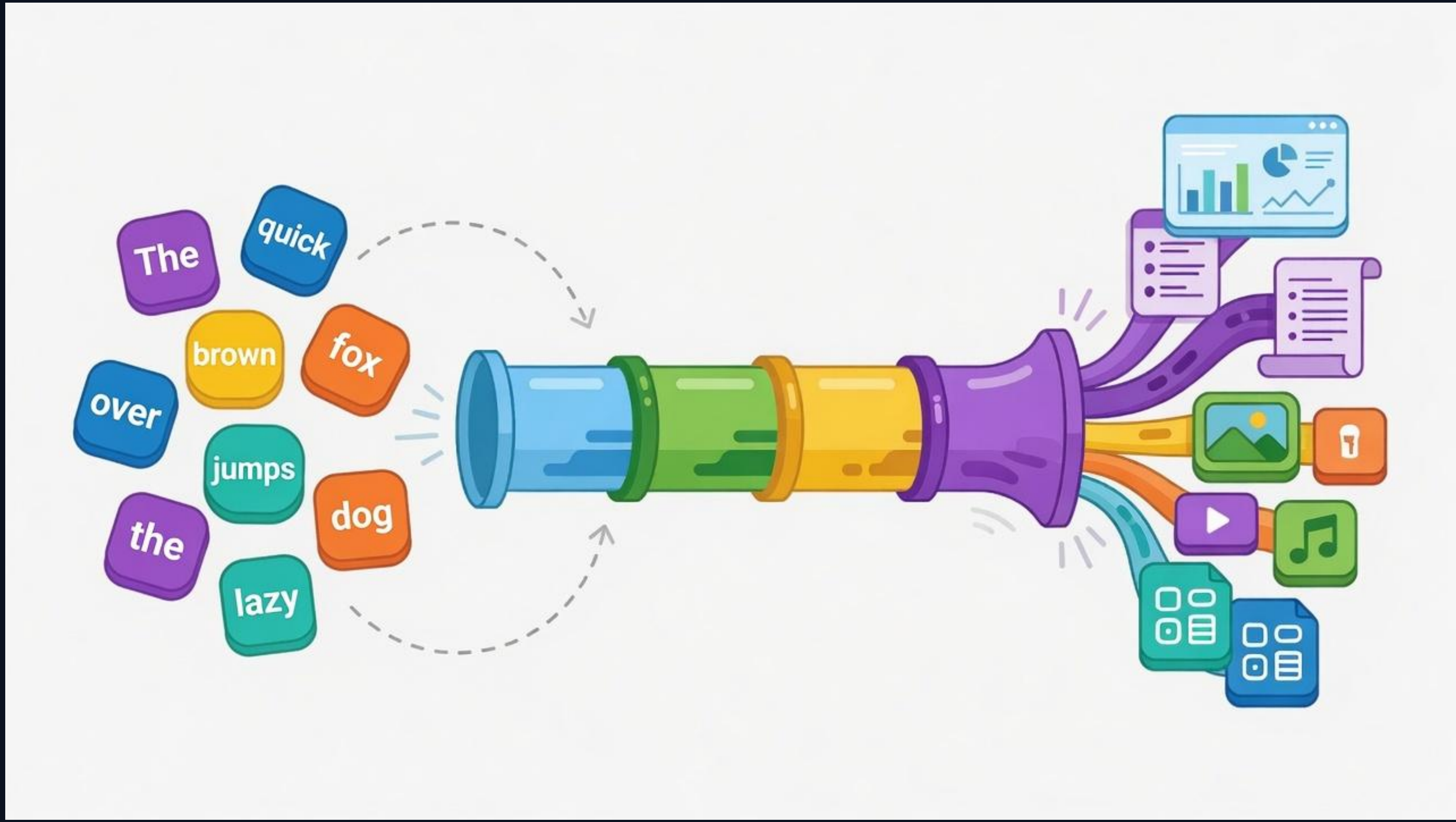
Token Efficiency



Claude
BY ANTHROPIC

INVOICE

DATE: AUGUST 2024
INVOICE # CLAUDE-08-2024

BILL TO:
VALUED CUSTOMER
[illegible]
[illegible]

DESCRIPTION	AMOUNT (USD)
Claude API Usage (July 1 – July 31, 2024) Usage-based charges for API access, models, compute, and related services.	**$500,000,000.00**

TOTAL DUE

$500,000,000.00

Thank you for building with Claude.
We appreciate your partnership.

SUBTOTAL	$500,000,000.00
TAX (0%)	$0.00
TOTAL DUE (USD)	$500,000,000.00

Anthropic PBC
548 Market St, PMB 97275
San Francisco, CA 94104
United States
billing@anthropic.com

PAYMENT TERMS
Due upon receipt.

ANNOUNCEMENT

Linux Foundation Announces the Intent to Launch the Tokenomics Foundation to Establish Open Standards for AI Cost Management

THE LINUX FOUNDATION

= 4 tokens


1. Cost is invisible during design
You only see it on the bill later.
ChatCompletion.create(
model: "gpt-4o",
messages: [
{ role: "system", content: ... },
{ role: "user", content: ... }
],
max_tokens: 1000
temperature: 0.2
),
What will this cost?
Your Invoice
April 1 – April 30
Total
$1,842.72

2. Three token types. Three wildly different prices.

Every call is a mix — and the rates change constantly.

	Input Tokens (you pay)	Cached Input Tokens (you pay less)	Output Tokens (you pay more)
This call	8,542	45,128	1,203
Price / 1K tokens (example)	$0.015	$0.0015	$0.060
Cost for this call	$0.128	$0.068	$0.072

Rates change (live)

3. Rates vary by model, region, and tier.

There's no single price to memorize.

Live pricing (per 1K tokens)

Model: All models

Region: All regions

Tier: All tiers

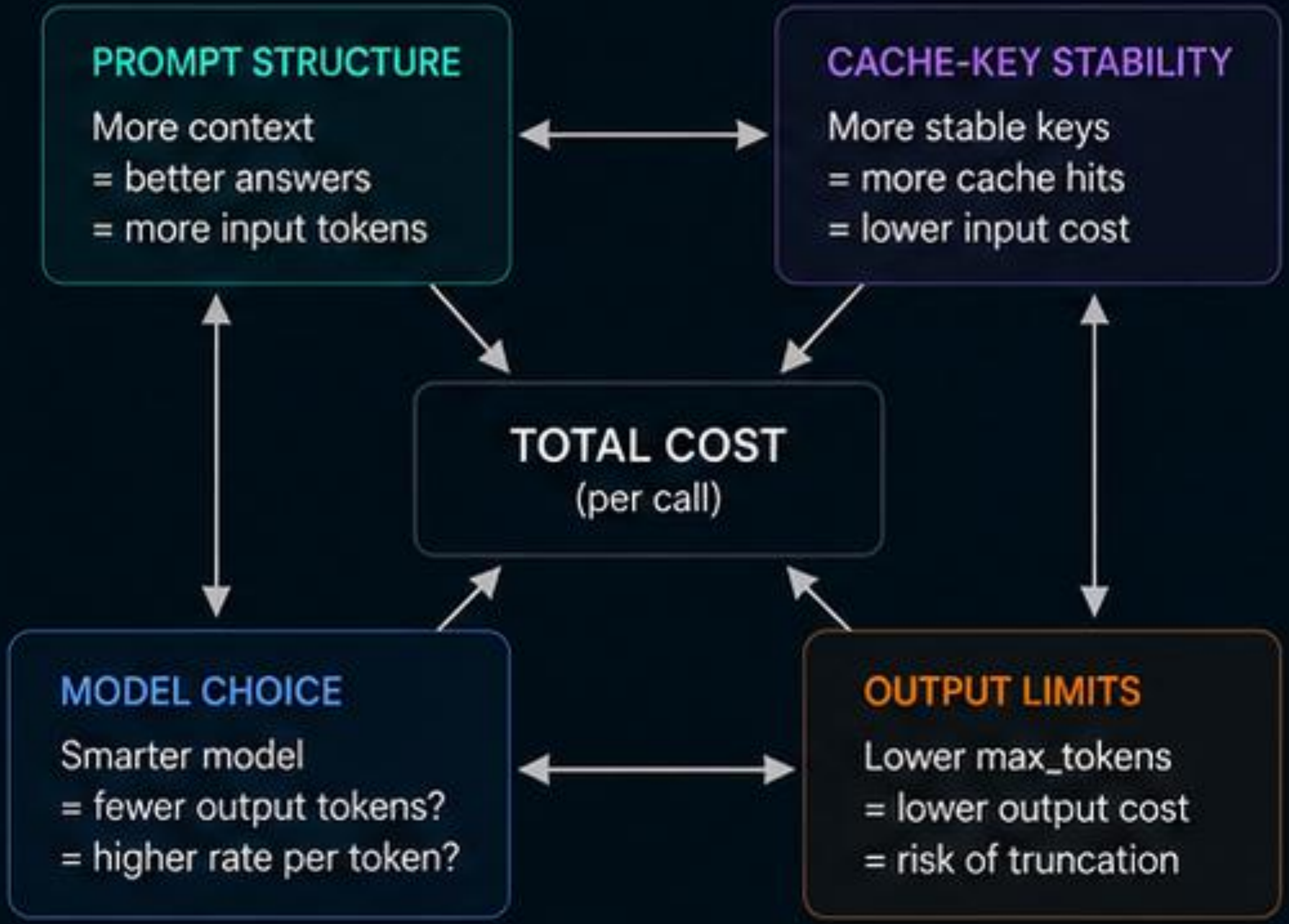
Model	Region	Tier	Input (You pay)	Cached Input (You pay less)	Output (You pay more)	Last updated
gpt-4o	East US	Standard	$0.015	$0.0015	$0.060	2 min ago
gpt-4o	East US	Provisioned	$0.010	$0.0010	$0.040	2 min ago
gpt-4o	Sweden Central	Standard	$0.016	$0.0016	$0.064	1 min ago
gpt-4.1	East US	Standard	$0.003	$0.0003	$0.012	1 min ago
gpt-4.1-mini	East US	Standard	$0.0008	$0.0001	$0.0032	1 min ago

4. You're juggling multiple interacting levers.

Change one thing, and everything shifts.

5. The levers interact in non-obvious ways.

Optimize one, and another might get worse.



6. You can't tune what you can't see.

You need per-call visibility to get it right.

Per-call token breakdown

Total cost	Input tokens	Cached input tokens	Output tokens
$0.2687	12,351 $0.1853	37,842 $0.0568	1,102 $0.0666

Breakdown

Type	Tokens	Rate / 1K	Cost	% of total
Input	12,351	$0.0150	$0.1853	69%
Cached input	37,842	$0.0015	$0.0568	21%
Output	1,102	$0.0605	$0.0666	25%
Total	**51,295**	—	**$0.2687**	**100%**

What-if simulation

Max output tokens	1000
Reduce input tokens by	-20%
Increase cache hit rate to	80%

Estimated new cost

$0.1762

↓ 34%

Project Setup

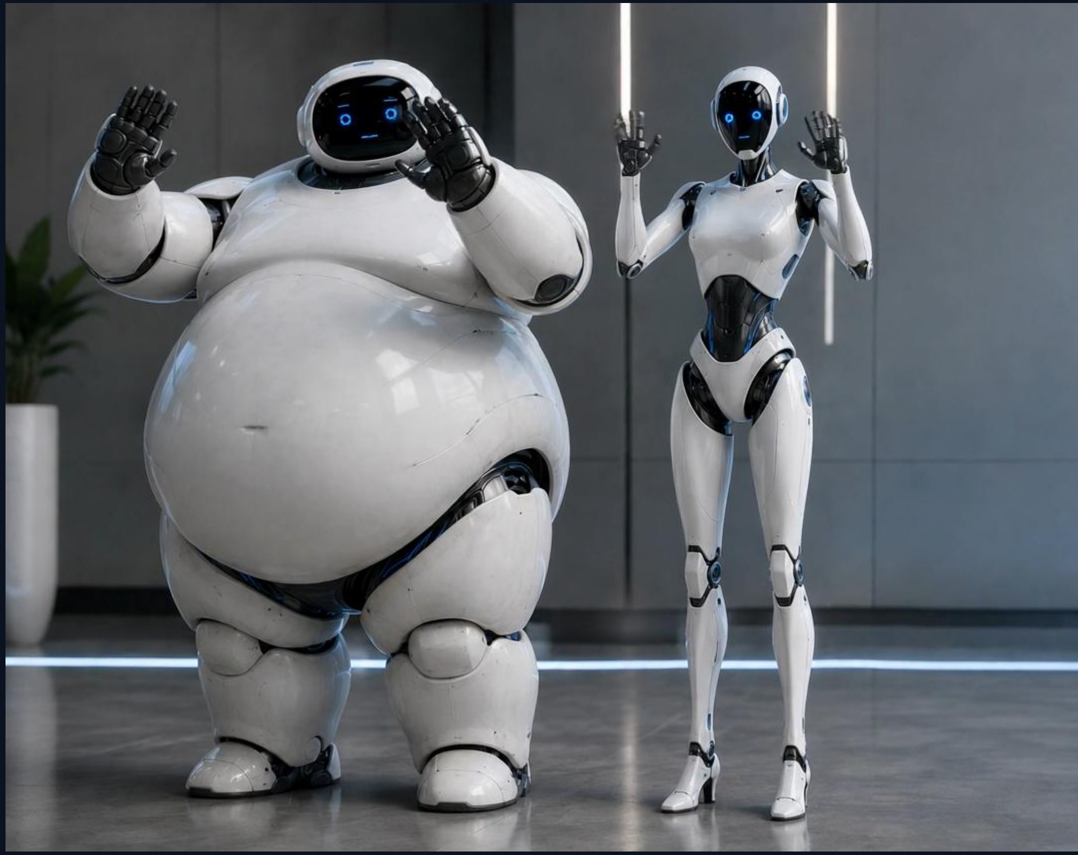
- devcontainer.json
- Copilot instructions
- **A**gent profile

Reduce input and context tokens

- Drop unused MCP tools
- Scope sessions tightly
- Keep prompts precise and short

Choose the right execution path

- Use the smallest model that fits
- Prefer a direct completion
- Watch output, not just input



MICROSOFT FOUNDRY JAVA SAMPLE

GitHub

Token Efficiency

Run an instant model call, compare prompt cache reuse, then compact long working notes into a smaller reusable prompt.

MODEL	METER	REGION
gpt-chat-latest	5.5 ShortCo	westus3 / Gl

LIVE PRICING

1 call

Instant Demo

An **instant model** is called by name — there's no deployment to create or manage, and it's available to every Foundry project automatically. It bills at standard pay-as-you-go rates, so it stays much cheaper than a data-zone or priority deployment. This call is priced live against Azure Retail Prices meters, then scaled to 1,000 and 1M runs.

Prompt

In one sentence, why are instant models useful for prototyping?

REUSABLE PREFIX

2 calls

Prompt Cache Demo

Watch the model cache warm up in real time. A warm-up call primes a long stable prefix, then the repeated call reuses it — see the cache fill, the cold→warm comparison, and exactly what was loaded.

Warm the cache

CONVERSATION HYGIENE

1 call

Compaction Demo

Long working notes pile up tokens on every turn. One call compacts them into a short, durable summary that keeps only the next-turn essentials — goal, key facts, and commands — so future prompts stay cheap. See the before→after token drop and reduction rate.

Working notes

Goal: ship a small Java dashboard change that explains instant model throughput limits.
Context gathered: Microsoft Foundry instant models are preview-scoped to
West US 3 and can be called by model

Pricing

USD
0.001025
COST · THIS CALL

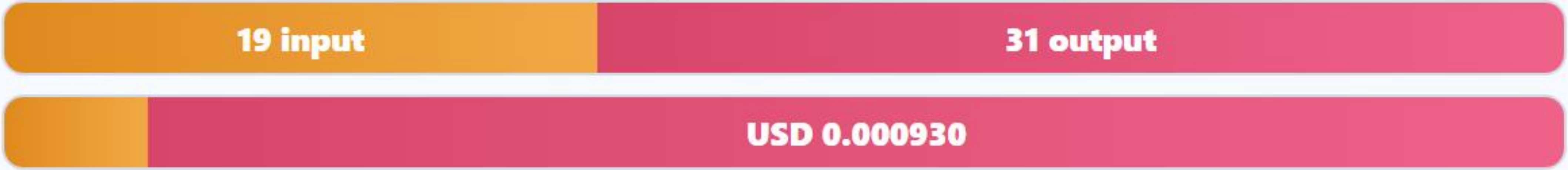
● **Live Azure retail rates**

USD 1.03
PER 1,000 CALLS

USD 1025.00
PER 1M CALLS

BY TOKEN COUNT

BY COST



Cached input **USD 0.00** Standard input **USD 0.00009500** Output **USD 0.000930**

Cooler = cheaper per token; warmer = pricier.

Prompt caching

96.86%
CACHE HIT ON REUSE

Cache warmed in real time
9,728 tokens served from cache

MODEL CACHE · GPT-CHAT-LATEST
CACHE WARM

9,728 cached
315 fresh

Cached prefix (warm)
Fresh input (cold)

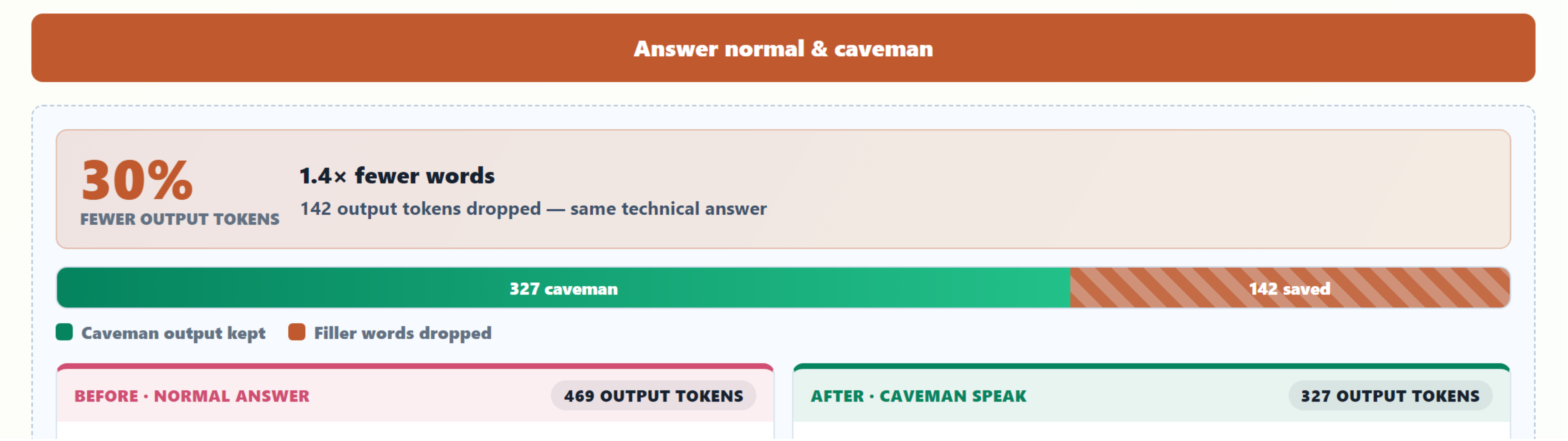
Compaction - Generic

37%
FEWER TOKENS

1.6× smaller

154 tokens reclaimed on every future reuse

Compaction – Cave man



Demo

https://aka.ms/costs